Practical MMC Noise Study Plan

Objective

Turn your uploaded MMC noise component files into a reconstruction-relevant study program that answers four questions:

1. **Which noise source dominates in which frequency band?**
2. **Which source actually limits baseline energy resolution?**
3. **Which frequency range matters most for pulse reconstruction and optimal filtering?**
4. **What part of the total noise is still unexplained or technical?**

This plan assumes you currently have frequency-domain spectra like: - `Er_noise.dat` - `Johnson_noise.dat` - `SQUID_noise.dat` - `TD_noise.dat` - `total_noise.dat` - `signal.dat`

Before starting: clarify the meaning of each file

Before doing any physics interpretation, write down the answer to these items in a notebook or README.

Required metadata

For each file, define: - Is the second column an **ASD**, **PSD**, **current noise**, **flux noise**, **power noise**, or a normalized spectrum? - What are the units? - Is `signal.dat`: - a pulse template in frequency domain, - a transfer function, - a responsivity, - or only a shape proportional to the signal? - Are all spectra one-sided or two-sided? - Are they already calibrated to deposited energy, or still in readout units?

Why this matters

All later studies depend on whether you should use: - `N(f)` directly, - `N^2(f)` as PSD, - or a transformed quantity such as NEP.

If this is not clarified first, the sensitivity study can become numerically consistent but physically wrong.

Study A — Noise budget decomposition

What this study means

This is the foundational study. It tells you: - which component dominates at low / mid / high frequency, - where crossover points occur, - whether your total noise is consistent with quadrature summation, - and which frequency bands are device-limited vs readout-limited.

This is the professional equivalent of saying: **what is the noise budget of the MMC?**

Physics question

For each frequency f, compare the contribution of each source to the total:

$$N_{\text{tot}}^2(f) \stackrel{?}{=} \sum_i N_i^2(f)$$

Then compute the fractional contribution:

$$R_i(f) = \frac{N_i^2(f)}{N_{\text{tot}}^2(f)}$$

This tells you what fraction of the variance comes from each source.

Practical steps

Step A1 — Load and validate the files

- Read all `.dat` files.
- Check the frequency arrays are identical.
- Confirm monotonic increasing frequency grid.
- Check for zeros, NaNs, negative values.

Step A2 — Verify the total-noise construction

- Reconstruct:

$$N_{\text{model}}(f) = \sqrt{N_{Er}^2 + N_{Johnson}^2 + N_{SQUID}^2 + N_{TD}^2}$$

- Compare `N_model(f)` with `total_noise.dat`.
- Plot the ratio:

$$N_{\text{total}}/N_{\text{model}}$$

- If this ratio is not close to 1, do not proceed until you understand why.

Step A3 — Plot all components together

Make a log-log plot with: - total noise, - each component, - optionally the signal curve.

This is the overview figure for presentations.

Step A4 — Compute fractional contributions

At each frequency compute: - $R_{Er}(f)$ - $R_{Johnson}(f)$ - $R_{SQUID}(f)$ - $R_{TD}(f)$

Plot them vs frequency.

Interpretation: - a curve near 1 means that component dominates that band, - multiple curves comparable means a mixed regime.

Step A5 — Find crossover frequencies

For each pair of components, solve approximately where:

$$N_i(f) = N_j(f)$$

This gives the transition points between dominance regimes.

Examples: - SQUID-to-Johnson crossover, - thermal-to-SQUID crossover, - excess low-frequency to white floor crossover.

Step A6 — Create a summary table

Produce a table with columns: - frequency band, - dominant noise source, - second-largest source, - fraction of total variance from dominant source, - comments.

Example:

Band	Dominant	Secondary	Dominant Fraction	Interpretation
1e0–1e3 Hz	SQUID	TD	80%	readout-dominated low-f
1e3–1e5 Hz	Johnson	SQUID	60%	white floor region
…	…	…	…	…

Output of Study A

- all-noise log-log plot,
- fractional contribution plot,
- crossover-frequency table,
- short written interpretation.

What conclusion you want

A good conclusion sounds like:

> Below frequency X, the noise is dominated by SQUID/readout. Between X and Y, Johnson noise sets the floor. Above Y, the thermal/device contribution becomes subdominant and the total spectrum follows the white readout level.

Study B — Resolution impact study

What this study means

A source can look large in a plot and still matter little for energy resolution if the signal has little weight there. This study answers the real question:

Which noise source hurts σ_E the most?

Physics question

For optimal filtering, sensitivity is governed by signal-to-noise weighting. In the simplest form, the information content scales like:

$$I \propto \int \frac{|S(f)|^2}{N_{\text{tot}}^2(f)}\, df$$

and therefore baseline energy uncertainty scales like:

$$\sigma_E \propto I^{-1/2}$$

Exact prefactors depend on how `signal.dat` is defined.

Practical steps

Step B1 — Define the signal quantity correctly

You must decide what `signal.dat` represents.

Cases: - If it is a frequency-domain pulse template proportional to response per unit energy, you can use it directly in OF weighting. - If it is only a relative shape, you can still do comparative studies, but not absolute σ_E in eV yet. - If it is not yet in the correct domain, convert it first.

Step B2 — Compute baseline sensitivity with all noise included

Calculate the OF information integral using total noise. This is your baseline performance.

Step B3 — Leave-one-out study

Recompute the total noise after removing one component at a time: - without SQUID, - without Johnson, - without TD, - without Er.

For each case, recompute the predicted σ_E.

Interpretation: - the component whose removal gives the biggest improvement is the main resolution bottleneck.

Step B4 — Scale-down study

Instead of total removal, scale each component by realistic factors: - 0.8x, - 0.5x, - 0.2x.

Then recompute σ_E.

This is much more useful for R&D decisions because "remove SQUID noise entirely" is usually not realistic, while "reduce SQUID noise by 2x" may be.

Step B5 — Rank the engineering priorities

Create a table:

Scenario	Predicted σ_E	Improvement vs baseline
Baseline	…	0%
Remove SQUID	…	…
SQUID × 0.5	…	…
Johnson × 0.5	…	…
TD × 0.5	…	…

Output of Study B

- baseline sensitivity value,
- leave-one-out table,
- scale-down table,
- ranking of which source is most worth improving.

What conclusion you want

A strong conclusion sounds like:

> Although SQUID noise dominates the low-frequency ASD, reducing Johnson noise yields less improvement than reducing SQUID noise because the pulse signal still carries significant OF weight in the SQUID-dominated band.

or

> TD noise is visible but contributes little to σ_E, so it is not the main threshold bottleneck.

Study C — Signal-band weighting study

What this study means

This study identifies the frequency region that actually matters for reconstructing your MMC pulse.

Noise far outside that region is visually dramatic but practically unimportant.

Physics question

Compute the information density:

$$W(f) = \frac{|S(f)|^2}{N_{\text{tot}}^2(f)}$$

This tells you where the optimal filter gets most of its discrimination power.

Then compute the cumulative information:

$$C(f) = \frac{\int_{f_{\min}}^{f} W(f')\,df'}{\int_{f_{\min}}^{f_{\max}} W(f')\,df'}$$

The frequencies where $C(f)$ passes 10%, 50%, 90% define the effective signal band.

Practical steps

Step C1 — Compute OF weighting density

Using your signal and total noise, compute $W(f)$. Plot it on log-log or semilog x axes.

Step C2 — Compute cumulative contribution

Integrate numerically to get $C(f)$. Plot cumulative fraction vs frequency.

Step C3 — Extract bandwidth landmarks

Find frequencies where cumulative information reaches: - 10% - 25% - 50% - 75% - 90%

This gives a compact summary of the frequencies that matter most.

Step C4 — Overlay component dominance

Compare Study C with Study A. For example: - if 80% of OF information lies in a band dominated by SQUID noise, that tells you SQUID optimization is high priority, - if the important band is mostly Johnson-limited, focus on thermal/readout floor reduction, - if useful information avoids the worst low-frequency region, then huge low-frequency excess may not be your main threshold problem.

Output of Study C

- OF weighting plot,
- cumulative information plot,
- effective signal band summary,
- cross-interpretation with noise budget.

What conclusion you want

A good conclusion sounds like:

> 90% of the useful OF information is concentrated between frequencies A and B, so noise reduction outside this band will have only marginal effect on threshold.

Study D — Excess-noise residual model

What this study means

This study asks whether the total noise is fully explained by your known component model. If not, the leftover structure is "excess noise," and you should classify it rather than hand-wave it.

Physics question

Define a residual between measured total and modeled known components. For PSD-like quantities, the cleanest residual is:

$$R(f) = N_{\rm meas}^2(f) - N_{\rm known}^2(f)$$

Then test whether the residual resembles one of these classes: - **1/f-like** drift or low-frequency instability, - **white excess** electronics floor, - **thermal-like rolloff** unmodeled sensor dynamics, - **line noise** narrow peaks from EMI or vibration, - **microphonic combs** or harmonics.

Practical steps

Step D1 — Build the known-noise model

Construct the model from your named components. If the uploaded `total_noise.dat` already equals the quadrature sum exactly, this study becomes relevant later when you compare the component model to actual measured baseline PSDs from data.

Step D2 — Compute the residual spectrum

Plot: - residual in linear scale where possible, - residual fractional ratio, - residual in log-log for broad-shape interpretation.

Step D3 — Fit simple residual models

Test models such as:

$$R(f) = A_{1/f} f^{-\alpha} + A_w$$

or

$$R(f) = A_{1/f} f^{-\alpha} + A_w + A_{th} H_{th}(f)$$

where $H_{th}(f)$ is a thermal-rolloff-like shape.

Step D4 — Detect line features

Automatically search for narrow peaks above a smoothed continuum. For each line record: - center frequency, - amplitude above baseline, - width, - possible source.

Typical suspects: - mains pickup, - harmonics, - pulse tube / mechanical resonance, - switching electronics, - clocks.

Step D5 — Classify the residual

Assign each major feature to a class: - broad low-f drift, - white excess, - thermal-like hump, - narrow lines, - unknown.

Output of Study D

- residual plot,
- residual fit parameters,
- list of identified line features,
- classification summary.

What conclusion you want

A strong conclusion sounds like:

> The modeled intrinsic noise explains the measured spectrum above frequency X, but a low-frequency excess consistent with $1/f^{1.2}$ and several narrow lines at Y and Z remain, indicating technical rather than intrinsic limitations.

Study E — Operating-point scan

What this study means

Noise and sensitivity depend strongly on working point. The best-looking PSD at one setting may not give the best energy resolution overall.

This study finds the best MMC operating condition.

Practical steps

Perform the same decomposition and sensitivity calculation at multiple settings: - different sensor bias points, - different temperatures, - different SQUID operating points, - different feedback/gain settings, - different magnetic/environmental settings if relevant.

For each setting, compute: - total PSD, - noise budget fractions, - predicted σ_E, - effective signal band.

Create summary plots: - σ_E vs operating point, - component amplitude vs operating point, - dominant source map vs operating point.

Output of Study E

- best operating point,
- sensitivity trend plots,
- explanation of what physically changes with operating point.

Study F — Technical noise diagnosis

What this study means

This is how you separate intrinsic detector noise from environmental or readout-chain noise.

Practical steps

Repeat baseline PSD measurements under controlled changes: - shielding changed / unchanged, - grounding changed / unchanged, - vibration source on / off, - pulse tube on / off if available, - different cabling or amplifier chain, - different bath conditions, - SQUID settings varied.

For each change, compare: - total PSD, - line amplitudes, - low-frequency floor, - predicted σ_E.

Output of Study F

- before/after PSD comparisons,
- attribution of specific excess features,
- list of likely technical bottlenecks.

A practical execution schedule

Phase 1 — Immediate work with your uploaded files

Do this now with the existing spectra.

1. Load all six files.
2. Verify frequency grids and units.
3. Confirm total-noise quadrature sum.
4. Make the all-components plot.
5. Compute fractional contributions.
6. Find crossover frequencies.
7. Compute OF weighting $|S|^2/N^2$.
8. Make cumulative information plot.
9. Run leave-one-out and scale-down sensitivity studies.
10. Write a 1-page summary of conclusions.

Phase 2 — Real detector-data validation

Do this when you have time-domain baseline traces.

1. Collect many clean random-trigger baselines.
2. Remove pulses/glitches/drifts.
3. Estimate measured PSDs.
4. Compare measured PSD to your component model.
5. Fit excess residuals.
6. Validate predicted σ_E against OF baseline fluctuations.

Phase 3 — Optimization and diagnosis

Do this after the first interpretation.

1. Repeat measurements across operating points.
2. Repeat under environmental changes.
3. Rank hardware and readout improvements by expected gain in σ_E.
4. Choose the best path for threshold improvement.

Minimal deliverables for a strong internal report

Figures

1. All noise components + total + signal.
2. Fractional contribution vs frequency.
3. OF weighting density.
4. Cumulative information curve.
5. Leave-one-out sensitivity bar plot.
6. Scale-down sensitivity plot.
7. Residual / excess-noise plot.
8. Before/after comparison plots for technical tests.

Tables

1. Crossover frequencies.
2. Dominant noise source by band.
3. Predicted σ_E by scenario.
4. Operating-point comparison table.
5. Technical-change comparison table.

What to do first, concretely

This week

- Complete Study A.
- Complete Study C.
- Start Study B with relative sensitivity if absolute calibration is not ready.

Next step after that

- Use measured baseline traces to validate or replace the synthetic/noise-component PSDs.
- Then move to excess-noise modeling and operating-point scans.

Common mistakes to avoid

1. **Using ASD and PSD interchangeably.**
2. **Doing energy-resolution calculations before signal units are understood.**
3. **Interpreting the visually largest noise feature as the most important one.**
4. **Ignoring narrow lines because they occupy little bandwidth; some can still hurt OF or timing.**
5. **Using one static PSD if the system is nonstationary.**

6. **Calling everything “intrinsic” before environmental tests are done.**

Recommended notebook structure

Notebook 1 — Data loading and validation

- read files
- unit checks
- quadrature-sum validation

Notebook 2 — Noise budget decomposition

- plots
- fractional contributions
- crossover extraction

Notebook 3 — Sensitivity study

- signal/noise weighting
- cumulative information
- leave-one-out
- scale-down study

Notebook 4 — Measured PSD comparison

- time-domain baseline selection
- PSD estimation
- residual modeling

Notebook 5 — Operating-point and technical comparisons

- repeated analyses
- comparison tables
- optimization summary

Final mindset

The purpose of these studies is not merely to plot noise curves. The purpose is to build a chain:

noise sources -> total spectrum -> useful signal band -> predicted energy sensitivity -> engineering priority

That chain is what turns raw MMC noise data into a detector-optimization result.